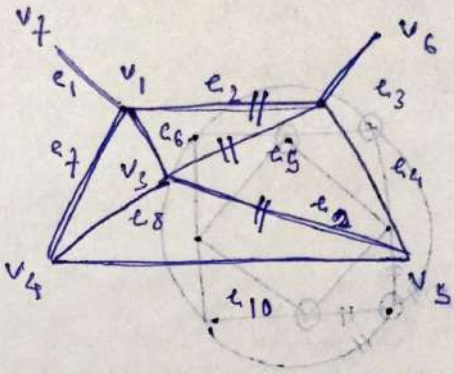
Graph Theory

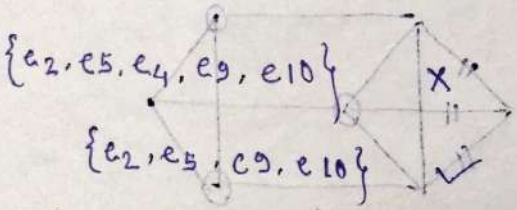
Cut-set (Edges):

In a connected graph G, a cut set is a set of edges whose removal from G leaves G disconnected provided removal of no proper subset of these edges disconnects G.

cut-set	validity
$\{e_2, e_5, e_9\}$	✗
$\{e_2, e_6, e_8, e_{10}\}$	✓
$\{e_1, e_2\}$	✗



Target - disconnected graph.



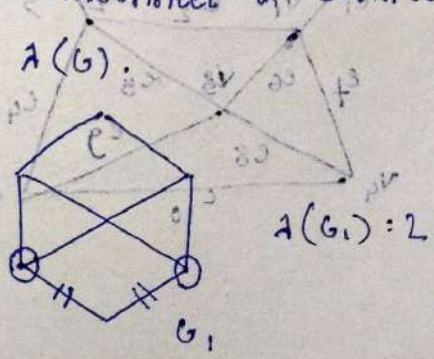
$\{e_6, e_5, e_9, e_8\}$ ✓

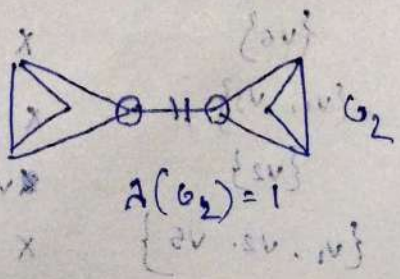
Note:

1. A cut set always cuts a graph into two.
2. Every edge of a tree is a cut set of a tree.
3. A cut set is a subgraph of G.

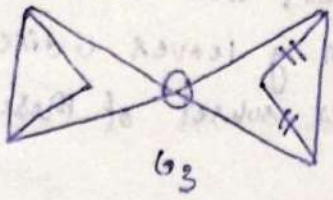
Edge connectivity:

The min no. of edges removed to disconnect a connected graph. denoted by $\lambda(G)$.



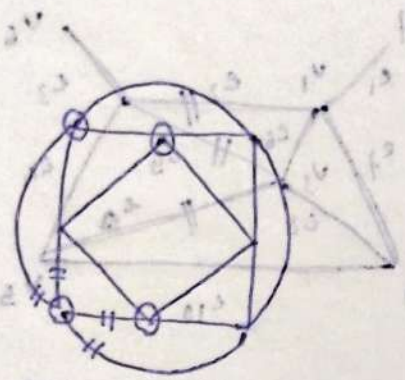


If in a connected graph there is an edge such that on removing it the graph becomes disconnected. then that edge is called bridge.



G_3

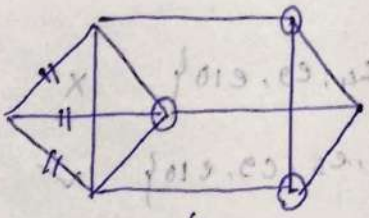
$\lambda(G_3) = 2$



G_4

$\lambda(G_4) = 4$

$\lambda(G_5) = 3$

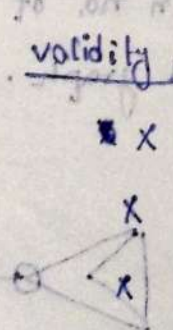


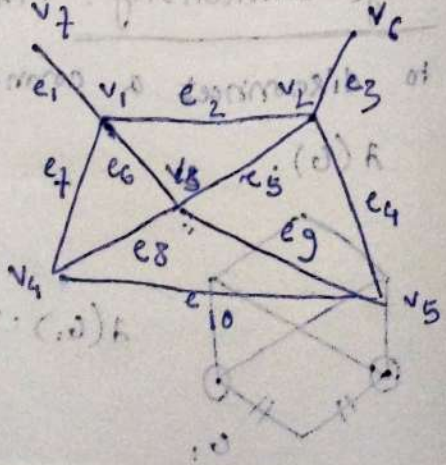
G_5

- **cut set (vertex):**

In a connected graph G, a cut set is a subset of vertices whose removal leaves G disconnected provided removal of no proper subset of these vertices disconnects G.

cut set	validity
$\{v_3, v_5\}$	X
$\{v_6\}$	X
$\{v_1, v_3\}$	X
$\{v_2\}$	✓
$\{v_1, v_2, v_5\}$	X





vertex connectivity: Each cut set of a connected graph G consist of a certain no. of vertices. The no. of vertices in the smallest cut set is defined as the vertex connectivity of G. It is denoted by $\kappa(G)$.

- A connected graph is said to be separable if its vertex connectivity is one.
- If the vertex connectivity of a graph is one, then that vertex whose removal disconnectes a graph is called articulation point.

Note:

1. A graph may have more than one articulation point.
2. Every vertex (with degree greater than one) of a tree is cut-vertex.
3. Definition of cut vertex can be extended to graph having several components also. There, a cut vertex would disconnect a component of a graph.

	G1	G2	G3	G4	G5
$\kappa(G)$	2	1	1	4	3
$\lambda(G)$	2	1	2	4	3
$\delta(G)$	2	2	2	4	3

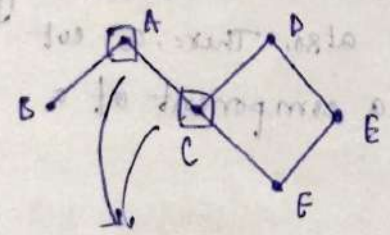
$$\kappa(G) \leq \lambda(G) \leq \delta(G).$$

■ **Separable graph & its block:** A connected graph (or a connected component of a graph) is said to be separable if it has a articulation point. A connected graph (or a connected component of a graph) which is not w.r.t a spanning tree.

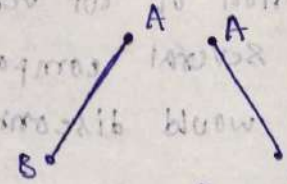
∴ A separable graph consists of two or more non-separable subgraphs.

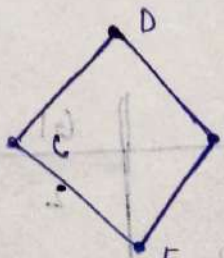
■ **Note:**

1. The vertex connectivity of tree is 1.
2. The above definition is appropriate for those graphs which have three or more vertices and are not complete.
3. The vertex connectivity of a connected separable graph is 1.



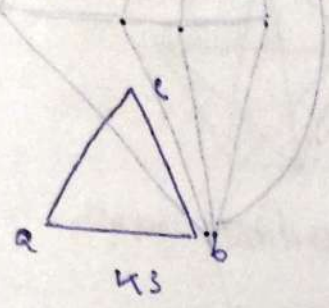
Articulation point.





Blocks.

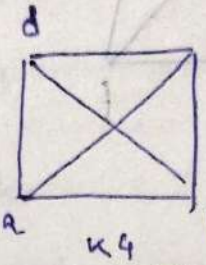
Planner Graph: A graph is called a planner graph if it can be drawn on a plan in such a way that no edges cross each other. otherwise it is called non-planner.



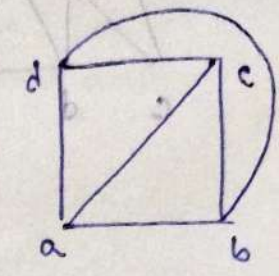
K_3

Planner graph.

complete graph with 3 vertices.

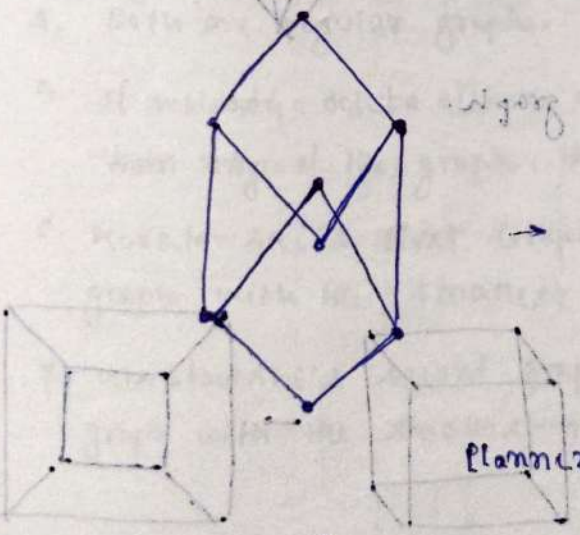


K_4

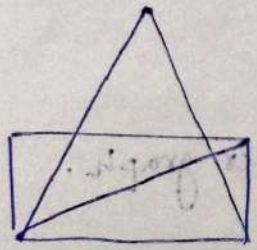


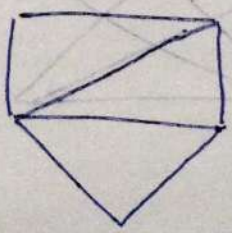
Planner graph.

Note: To declare a graph as a planner we must see wheather it can be drawn on a plane without a crossover between its edge.

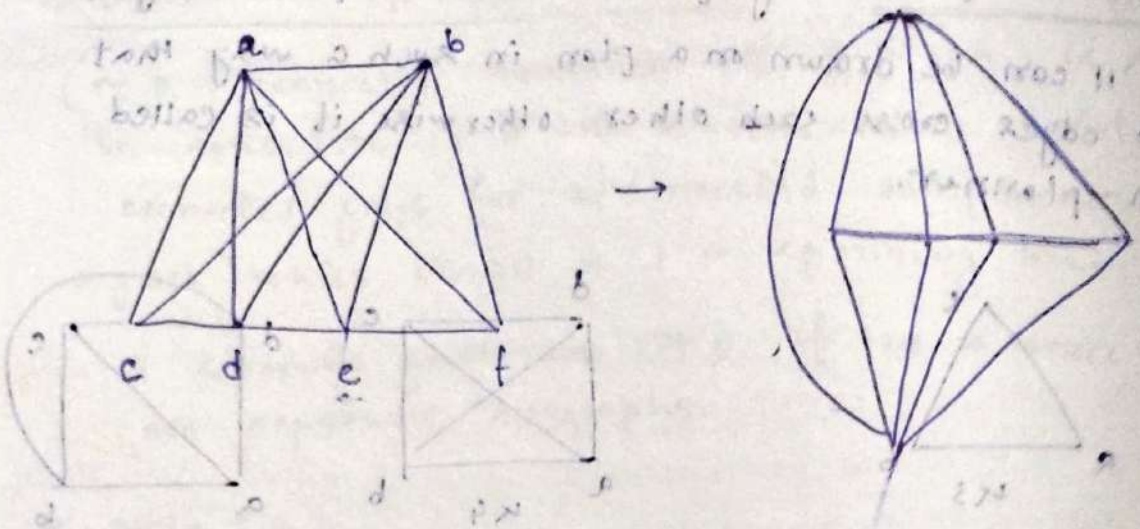


Planner graph.

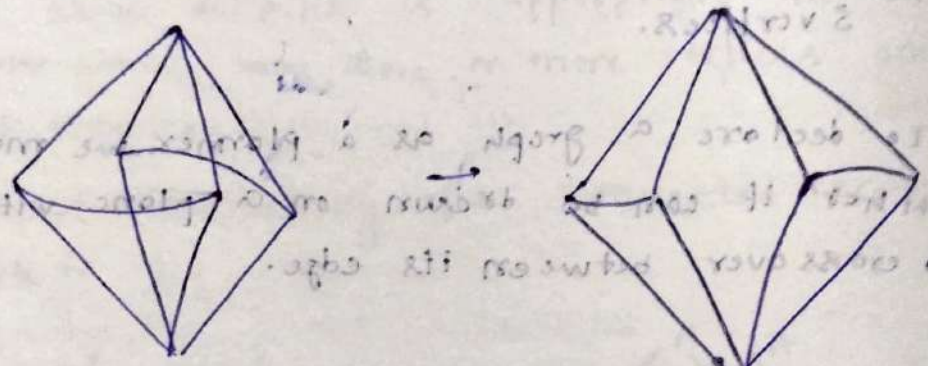




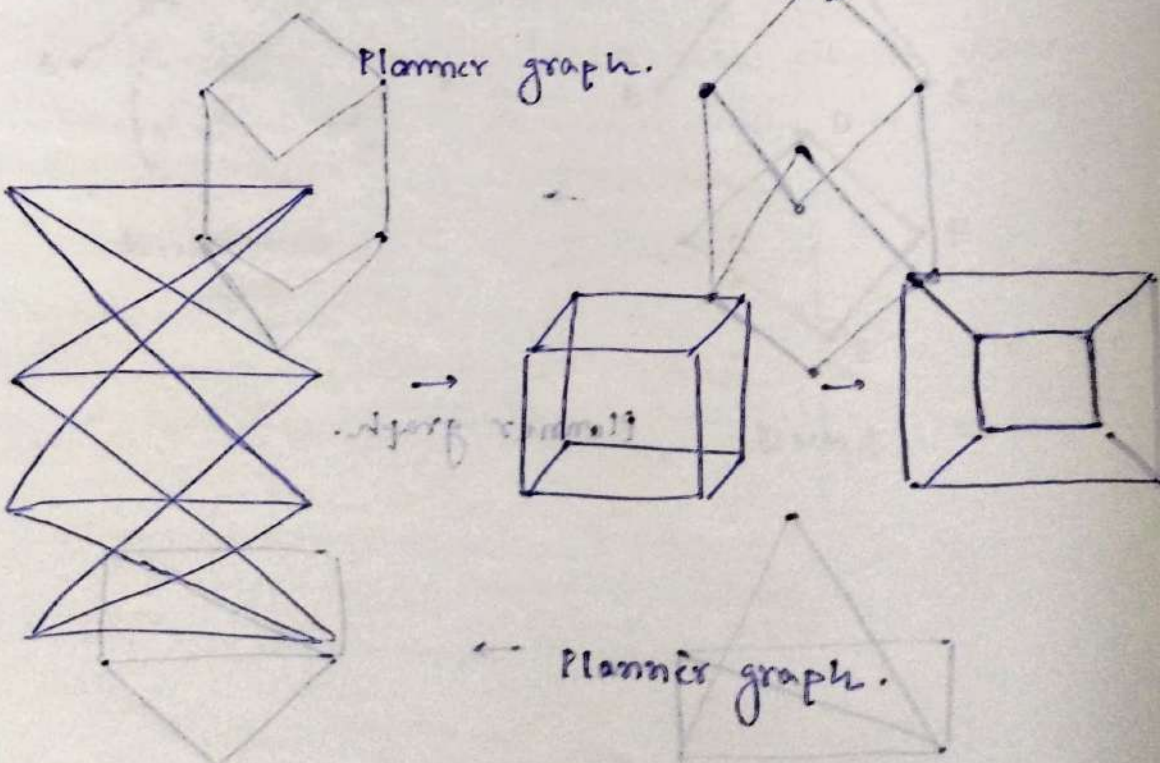
Planner graph.



Planner graph.

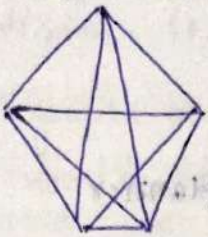


Planner graph.

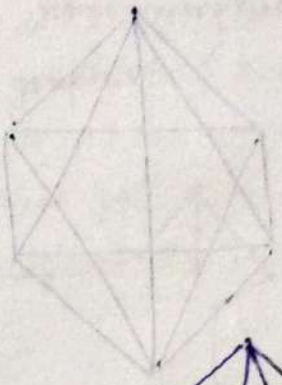


Planner graph.

- **Non-planner graph**: A graph which is not planner. is called non-planner graph.

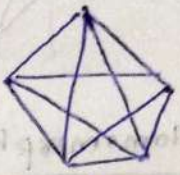


Non-planner graph.



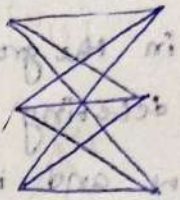
- **Simplest Non-planner graph**:

1. Kuratowski's first graph - K_5



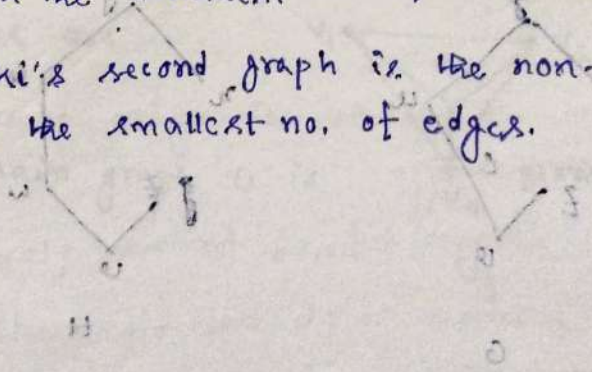
K_5

2. Kuratowski's second graph - $K_{3,3}$
3. Both are simple non-planner graph.
4. Both are regular graph.



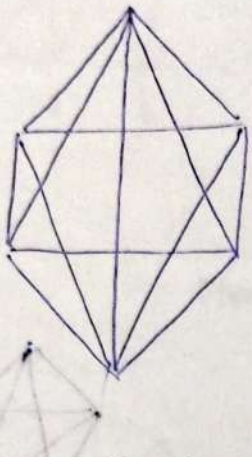
$K_{3,3}$

5. If we delete either an edge or vertex from any of the graph, they will become planner.
6. Kuratowski's First Graph is the non-planner graph with the smallest no. of vertices.
7. Kuratowski's second graph is the non-planner graph with the smallest no. of edges.



Kuratowski's Theorem:

A graph G is planner if and only if G does not contain either of two Kuratowski's two graphs.



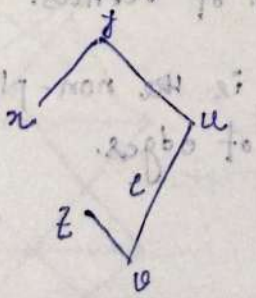


Non-planner

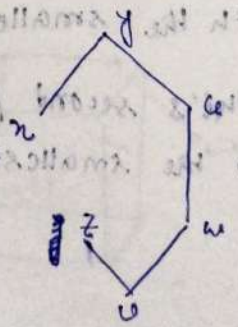
contains first graph.

Homomorphic Graph:

Let e be an edge joining the two vertices u and v in the graph G. Let a new graph H is formed by deleting the edge e and introducing a new vertex w and introducing a new edges. one joining u and w and other joining v and w. This operation of replacement of an edge by two edges and a new vertex is called edge subdivision.



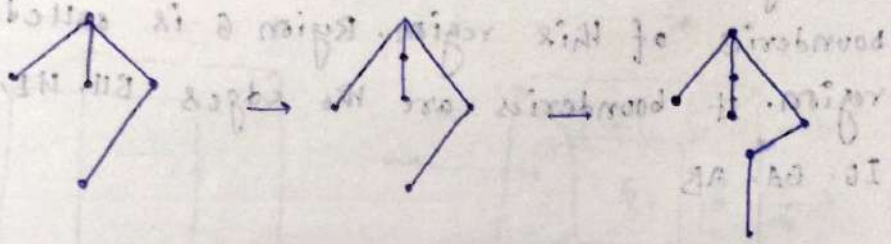
G



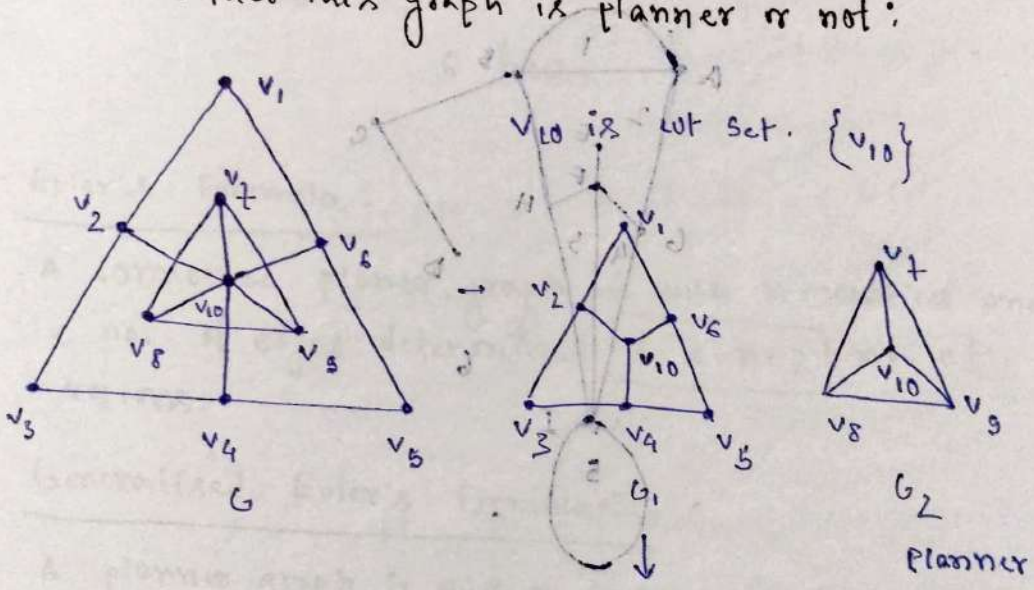
H

A graph obtained from a graph G by a sequence of edge subdivision, is called Homeomorph (or a subdivision) of G. (In figure H is obtained from G by only one edge subdivision. so, H is Homeomorph of G.

Two graphs are said to be homeomorphic if each is a homeomorph of same graph.

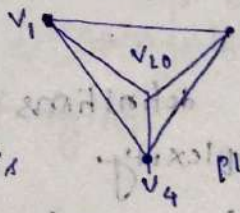


check wheather this graph is planner or not:



Both blocks are planner.

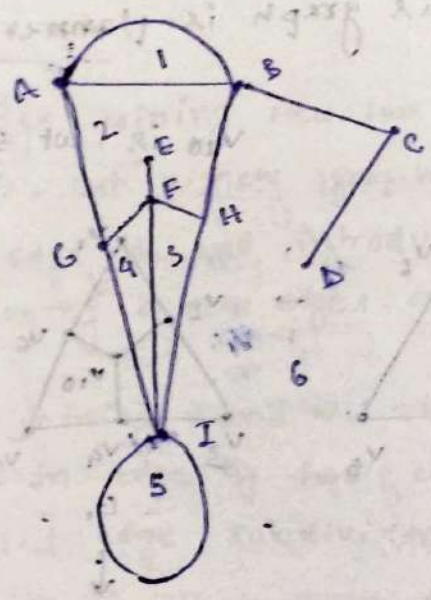
so, the main graph G is planner.



Region (or. Face) of a planner graph :

A plane graph divides the graph into regions.

The regions of the graph are shown by the numbers 1, 2, 3 ... eg. Region 2 is determined (or. bounded) by the edges AB, BH, HF, FG and GA. we say AB, BH, HF, FG and GA are bounded boundaries of this region. Region 6 is called infinite region. Its boundaries are the edges BH, HF, FI, IG, GA, AB

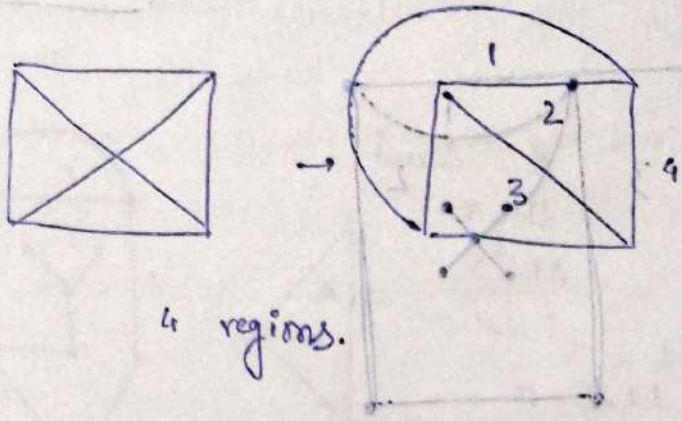


Note :

1. The formal definitions of region is not given here to its complexity.
2. A region is not defined for non-planner graph.
3. Even a planner graph not represented as plane graph do not determine region.
4. By changing the drawing of a planner graph. we can change.

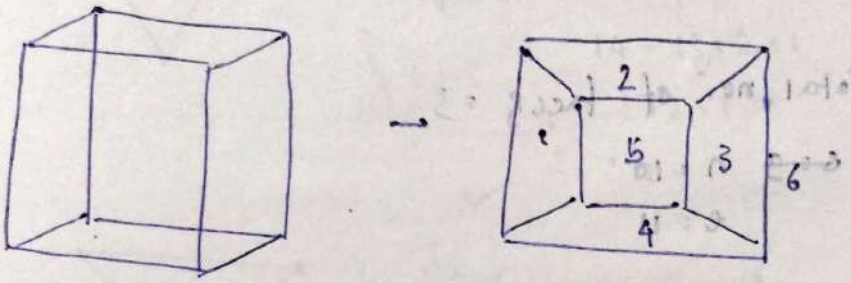
No. of faces of the planner Graphs:

1.



4 regions.

2.



6 faces.

- **Euler's Formula:**

A connected planer graph G with n vertices and e no. of edges determines $\boxed{f = e - n + 2}$ no. of regions.

- **Generalised Euler's Formula:**

A planner graph G with n vertices, e no. of edges and k no. of connected components determines $\boxed{f = e - n + k + 1}$ no. of region.

- **Theorem:**

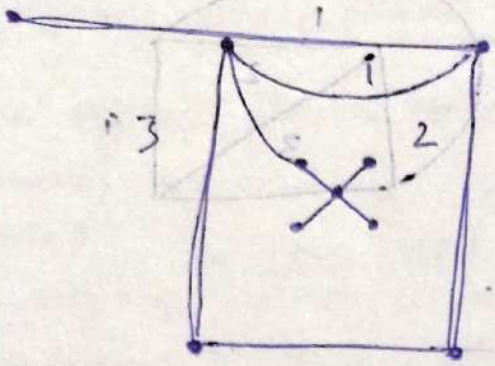
Let, G be a simple connected planer which with n vertices, e edges and f regions then

(i) $e \geq 3/2 f$

(ii) $e \leq 3n - 6$.

Verify Euler's Formula: $f = e - n + 2$

1.

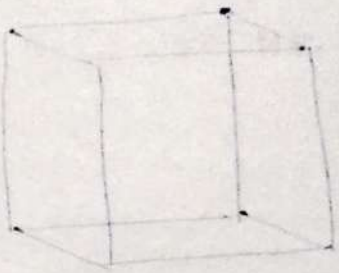




Total no. of faces = 3

$n = 10$

$e = 11$



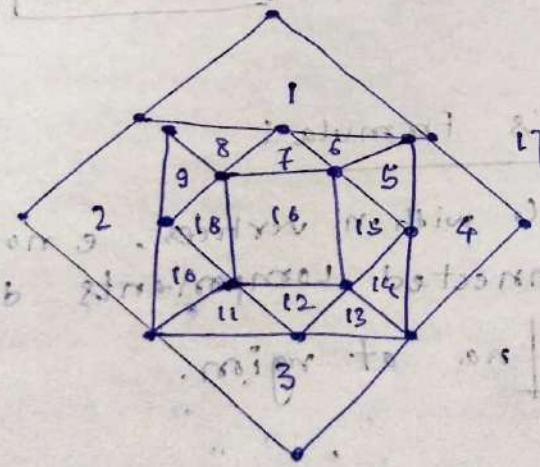
$\therefore e - n + 2 = 11 - 10 + 2$

$= 3$

$\therefore$ Total no of faces : $e - n + 2 = 3$

So, Euler's formula is verified.

2.



17

$\therefore$ Total no. of faces = 18

$n = 16$

$e = 32$

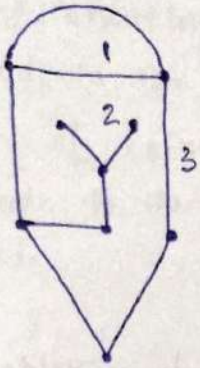
$\therefore e - n + 2 = 32 - 16 + 2$

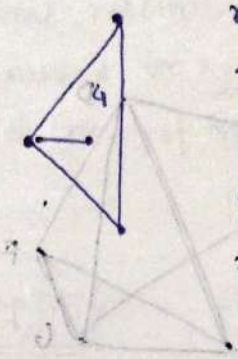
$= 16 + 2$

$= 18.$

$f = e - n + k + 1$

1.





faces = 4

$n = 13$

$e = 14$

$k = 2$

$e - n + k + 1$

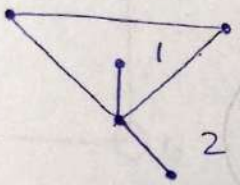
$= 14 - 13 + 2 + 1$

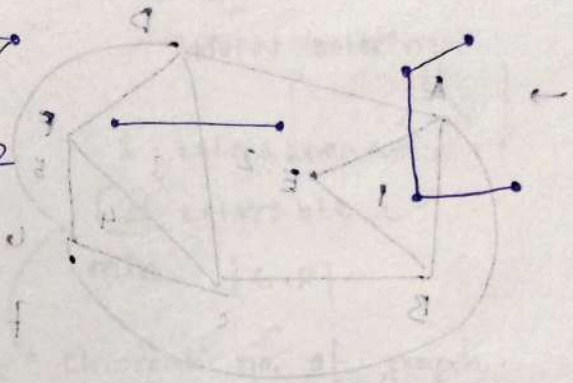
$= 1 + 2 = 1$

$= 4.$

✓

2.





Faces = 2

$e = 9$

$n = 11$

$k = 3$

$e - n + k + 1 = 9 - 11 + 3 + 1$

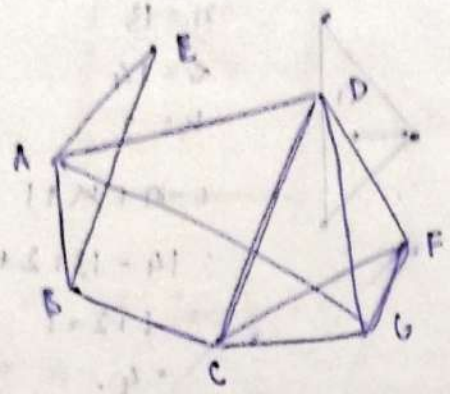
$= -2 + 3 + 1$

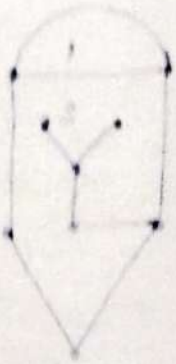
$= 2.$

✓

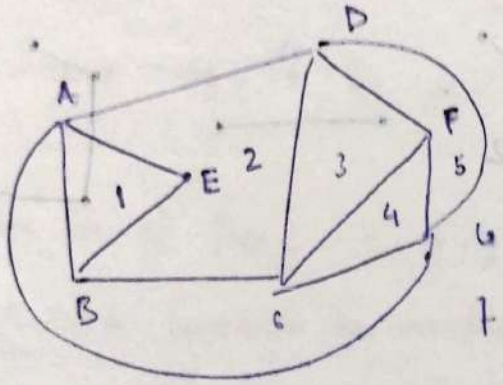
11. So redrawing the following graphs as plane graph show that they are planner : Find the no. of regions determined by each. Show the regions determined by these graph.

1.





→





Faces : 7

e : 12

n : 7

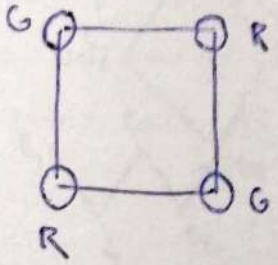
e - n + 2 = 12 - 7 + 2

= 5 + 2 = 7 no. of faces.

Graph Colouring

- **Scheduling**: Graph coloring is used to schedule tasks or events, ensuring that conflicting tasks are not assigned to the same resource or time slot. The min. no. of colors needed for a proper coloring corresponds to the min. no. of resources or time slots required.

- **vertex coloring of graphs**:



4 vertex, – 4 color

Trivial coloring.

3 colorable ✓

2-colorable ✓

min {G, R}

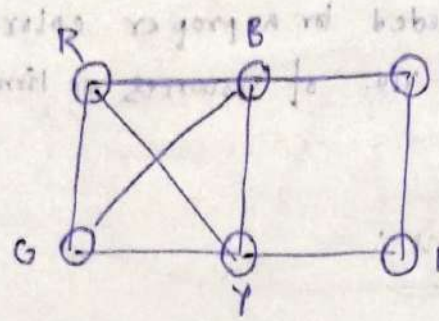
Chromatic no. of graph.

$\chi(G) = 2$

Associating a color with each vertex of the graph is called vertex coloring. By default graph coloring means vertex coloring even when simply coloring means vertex coloring.

A simple graph is said to be n-vertex colourable if it is possible to assign one color from a set of n-number of colors to each vertex such that no two adjacent vertices are assigned the same colour. We say the graph is coloured.

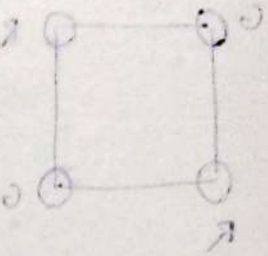
If a simple graph G is k vertex colourable but not (k-1) vertex colourable we say the graph G is a k-chromatic graph and that it chromatic number $\chi(G)$ is k.

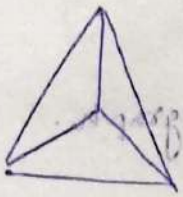


Trivial coloring - 6 colors.

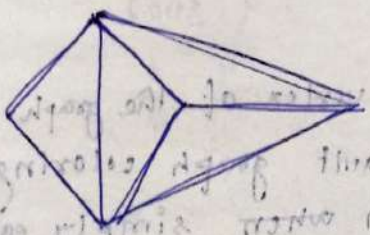
4-colorable

$\chi(G) = 4$

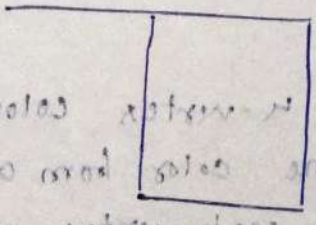




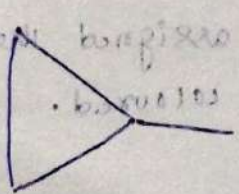
$\chi(G) = 4$



$\chi(G) = 4$

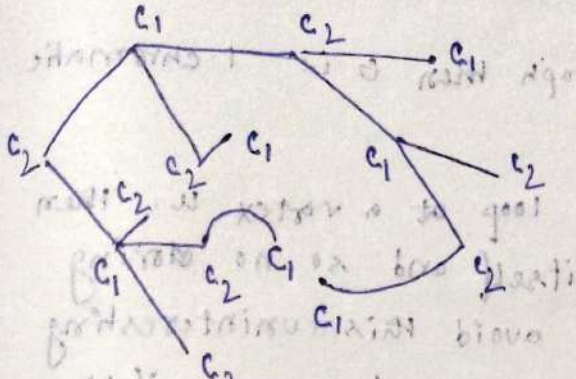


$\chi(G) = 2$

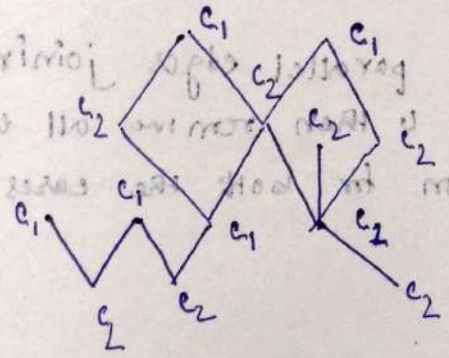


$\chi(G) = 3$

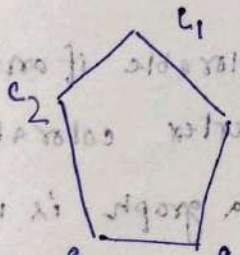
chromatic no. of tree is 2 always.



$\chi(G) = 2$



$\chi(G) = 2$



$\chi(G) = 3$

Note:

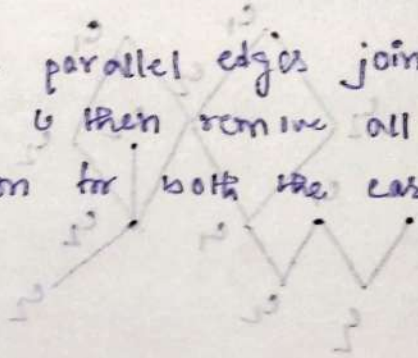
1. If $\chi(G) = k$ then k is the min. no. such that G is k-vertex-colourable.
2. A k-chromatic graph is a graph that needs at least k colors.
3. A k-vertex-colourable graph is a graph that does not need more than k-colors for its coloring. If the graph G has n no. of vertices then $\chi(G) \leq n$

4. If G is a single vertex graph then G is 1- chromatic graph.

5. If G is a null graph then G is 1 chromatic graph also.

6. If a graph G has a loop at a vertex u. Then u is adjacent to itself and so no coloring of G is possible. To avoid this uninteresting case we remove the loops of a graph if it has so.

7. If there exist some parallel edges joining two vertices of graph G then remove all except one because solution for both the cases is same.



Theorem:

1. A graph is K-vertex colorable if and only if each block in it is K-vertex colorable.

2. The chromatic no. of a graph is K if the chromatic no. of each block of it is K.

3. The chromatic no. of a complete ~~bipar~~ graph with n vertices (K_n) is n.